# Lie Algebras, Representations and Dynkin Diagrams

Jesse Vogel

## 1 Lie algebras

**Definition 1.1.** A **Lie algebra** is is a vector space $\mathfrak{g}$ over a field $k$, together with an operation $[\cdot,\cdot] : \mathfrak{g} \times \mathfrak{g} \to \mathfrak{g}, (x, y) \mapsto [x, y]$, called the **Lie bracket**, satisfying

- (*bilinearity*) $[ax + by, z] = a[x, z] + b[y, z]$ and $[z, ax + by] = a[z, x] + b[z, y]$ for all $x, y, z \in \mathfrak{g}$ and $a, b \in k$,
- (*alternativity*) $[x, x] = 0$ for all $x \in \mathfrak{g}$,
- (*Jacobi identity*) $[x, [y, z]] + [z, [x, y]] + [y, [z, x]] = 0$ for all $x, y, z \in \mathfrak{g}$.

**Example 1.2.** Most Lie algebras we will discuss arise from Lie groups. The Lie algebra corresponding to $\mathrm{GL}_n(k)$ is $\mathfrak{gl}_n(k) = \mathrm{Mat}_{n\times n}(k)$ with Lie brakcet $[X, Y] = XY - YX$. Furthermore, we have

$$\begin{aligned}
\mathfrak{sl}_n(k) &= \{X \in \mathfrak{gl}_n(k) : \mathrm{tr}(X) = 0\},\\
\mathfrak{so}_n(k) &= \{X \in \mathfrak{gl}_n(k) : X + X^T = 0\},\\
\mathfrak{su}_n(\mathbb{C}) &= \{X \in \mathfrak{gl}_n(\mathbb{C}) : \mathrm{tr}(X) = 0 \text{ and } X + X^H = 0\},\\
\mathfrak{sp}_{2n}(k) &= \{X \in \mathfrak{gl}_n(k) : \Omega X + X^T\Omega = 0\}, \text{ where } \Omega = \left(\begin{smallmatrix} 0 & I_n \\ -I_n & 0 \end{smallmatrix}\right).
\end{aligned}$$

For $\mathrm{char}(k) = 0$, it is not too restrictive to think of Lie algebras as subalgebras of $\mathfrak{gl}_n(k)$, because of the following theorem. We will not prove this theorem since we will not be needing it.

**Theorem 1.3** (Ado's Theorem)**.** *Every finite-dimensional Lie algebra over a field $k$ of characteristic zero is a subalgebra of $\mathfrak{gl}_n(k)$ for some $n$.*

**Definition 1.4.** Let $e_1, \ldots, e_r$ be a basis of the underlying vector space of a Lie algebra $\mathfrak{g}$. The Lie bracket can be described by coefficients $c_{ij}^k \in k$ such that $[e_i, e_j] = c_{ij}^k e_k$, called the **structure constants**. They satisfy the following relations:

$$c_{ij}^k + c_{ji}^k = 0 \quad (\textit{alternatively})$$

$$\sum_m c_{ij}^m c_{mk}^\ell + c_{jk}^m c_{mi}^\ell + c_{ki}^m c_{mj}^\ell = 0 \quad (\textit{Jacobi identity})$$

**Definition 1.5.** Let $\mathfrak{g}$ be a Lie algebra. A subspace $\mathfrak{h} \subset \mathfrak{g}$ is a **subalgebra** if $[h_1, h_2] \in \mathfrak{h}$ for all $h_1, h_2 \in \mathfrak{h}$, and $\mathfrak{h}$ is an **ideal** if $[g, h] \in \mathfrak{h}$ for all $g \in \mathfrak{g}$ and $h \in \mathfrak{h}$.

**Example 1.6** (Adjoint representation)**.** Every Lie algebra $\mathfrak{g}$ has a representation $\mathrm{ad} : \mathfrak{g} \to \mathfrak{gl}(\mathfrak{g})$ on itself, given by $\mathrm{ad}_X = [X, -]$. Indeed this is a representation since for all $X, Y \in \mathfrak{g}$,

$$\mathrm{ad}([X, Y]) = [[X, Y], -] = [X, [Y, -]] - [Y, [X, -]] = \mathrm{ad}(X)\mathrm{ad}(Y) - \mathrm{ad}(Y)\mathrm{ad}(X)$$

by the Jacobi identity and anti-symmetry.

## 2 Representation theory of $\mathfrak{sl}_2(\mathbb{C})$

The general representation theory of Lie algebras is best understood by first treating $\mathfrak{sl}_2(\mathbb{C})$ as an example. Consider the Lie algebra $\mathfrak{sl}_2(\mathbb{C})$, which has a basis given by

$$E = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}, \quad F = \begin{pmatrix} 0 & 0 \\ 1 & 0 \end{pmatrix}, \quad H = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix},$$

and commutations relations

$$[H, E] = 2E, \quad [H, F] = -2F, \quad [E, F] = H.$$

We want to classify the (irreducible) representations of $\mathfrak{sl}_2(\mathbb{C})$, so let $\pi : \mathfrak{sl}_2(\mathbb{C}) \to \mathfrak{gl}(V)$ be an irreducible representation. Let $v \in V$ be an eigenvector of $\pi(H)$ with eigenvalue $\lambda \in \mathbb{C}$. Then we see that

$$\pi(H)\pi(E)v = \pi(E)\pi(H)v + \pi([H, E])v = (\lambda + 2)\pi(E)v,$$

and similarly

$$\pi(H)\pi(F)v = \pi(F)\pi(H)v + \pi([H, F])v = (\lambda - 2)\pi(F)v.$$

This shows that $\pi(E)v$ and $\pi(F)v$ are either zero, or they are also eigenvectors of $\pi(H)$, with respective eigenvalues $\lambda \pm 2$. Repeating this argument, we find $\pi(H)\pi(E)^k v = (\lambda + 2k)\pi(E)v$ for all $k \geq 0$, and similarly for $F$. Since $\pi(H)$ only can have finitely many eigenvalues, there must exist some $N \geq 0$ such that $\pi(E)^N v \neq 0$ but $\pi(E)^{N+1} v = 0$. Set $v_0 = \pi(E)^N v$ and $\mu = \lambda + 2N$ and $v_k = \pi(F)^k v_0$ for $k \geq 0$. From the above we find

$$\pi(H)v_k = (\mu - 2k)v_k.$$

By induction, it is easy to find that $\pi(E)v_k = k(\mu - (k-1))v_{k-1}$ for $k \geq 1$. Since $\pi(H)$ only has finitely many eigenvalues, there must exist some integer $m \geq 0$ such that $v_k \neq 0$ for all $k \leq m$ but $v_{m+1} = 0$. In particular,

$$0 = \pi(E)v_{m+1} = (m+1)(\mu - m)v_m,$$

which implies $\mu = m$.

In conclusion, for every irreducible representation $\pi : \mathfrak{sl}_2(\mathbb{C}) \to \mathfrak{gl}(V)$, there exists an integer $m \geq 0$ and vectors $v_0, \ldots, v_m$ such that

$$\begin{aligned}
\pi(H)v_k &= (m - 2k)v_k, \\
\pi(E)v_k &= \begin{cases} k(m - (k-1))v_{k-1} & \text{if } k > 0 \\ 0 & \text{if } k = 0, \end{cases} \\
\pi(F)v_k &= \begin{cases} v_{k+1} & \text{if } k < m \\ 0 & \text{if } k = m. \end{cases}
\end{aligned}$$

The vectors $v_k$ are linearly independent as their corresponding eigenvalues are distinct. Moreover, from the above explicit expressions, the span of $v_0, \ldots, v_m$ yields an invariant subspace of $V$, and since $\pi$ is irreducible, this span must be the whole of $V$. This shows that every irreducible representation of $\mathfrak{sl}_2(\mathbb{C})$ is of this form, for some integer $m \geq 0$.

**Remark 2.1.** The irreducible representations can be made even more explicit. Take $V_m = \mathbb{C}[x,y]_m$ and define

$$\pi_m(E) = -y\partial_x, \quad \pi_m(F) = -x\partial_y, \quad \pi_m(H) = -x\partial_x + y\partial_y.$$

**Remark 2.2.** The fundamental representation $\pi : \mathfrak{sl}_2(\mathbb{C}) \to \mathfrak{gl}_2(\mathbb{C})$ corresponds to the case $m = 1$.

# 3 Solvable and nilpotent Lie algebras

**Definition 3.1.** A Lie algebra $\mathfrak{g}$ is called **nilpotent** if its lower central series

$$\mathfrak{g} \supset [\mathfrak{g}, \mathfrak{g}] \supset [\mathfrak{g}, [\mathfrak{g}, \mathfrak{g}]] \supset [\mathfrak{g}, [\mathfrak{g}, [\mathfrak{g}, \mathfrak{g}]]] \supset \cdots$$

terminates in the zero algebra. We also write $\mathfrak{g}^1 = \mathfrak{g}, \mathfrak{g}^2 = [\mathfrak{g}, \mathfrak{g}]$, etc.

**Definition 3.2.** A Lie algebra $\mathfrak{g}$ is called **solvable** if its derived series

$$\mathfrak{g} = \mathfrak{g}^{(0)} \supset \mathfrak{g}^{(1)} \supset \mathfrak{g}^{(2)} \supset \cdots$$

given by $\mathfrak{g}^{(i+1)} = [\mathfrak{g}^{(i)}, \mathfrak{g}^{(i)}]$ terminates in the zero algebra.

**Remark 3.3.** Note that any nilpotent Lie algebra is solvable, but not conversely. Let $\mathfrak{g} = \langle a, b \rangle$ with $[a, b] = b$, then $\mathfrak{g}^{(2)} = 0$ while $\mathfrak{g}^i = \langle b \rangle$ for all $i > 0$.

**Theorem 3.4** (Engel's theorem)**.** *Let $V$ be a non-zero vector space, and $\mathfrak{g} \subset \mathfrak{gl}(V)$ a finite-dimensional Lie subalgebra consisting of nilpotent operators. Then there exists a non-zero vector $v \in V$ such that $gv = 0$ for all $g \in \mathfrak{g}$.*

*Proof.* By induction on $\dim \mathfrak{g}$. If $\dim \mathfrak{g} = 1$, then $\mathfrak{g} = \langle g \rangle$ for some $g \in \mathfrak{g}$. Let $N > 0$ be such that $g^N \neq 0$ and $g^{N+1} = 0$, then any non-zero vector $v \in g^N V \neq 0$ satisfies $\mathfrak{g}v = 0$ by construction.

Now assume $\dim \mathfrak{g} \geq 2$. Let $\mathfrak{h} \subset \mathfrak{g}$ be a maximal proper subalgebra, which exists as $\mathfrak{g}$ is finite-dimensional. Since $\mathfrak{h}$ is a subalgebra, the adjoint representation induces a representation $\pi : \mathfrak{h} \to \mathfrak{gl}(\mathfrak{g}/\mathfrak{h})$. Now $\dim \pi(\mathfrak{h}) \leq \dim \mathfrak{h} < \dim \mathfrak{g}$ and $\pi(\mathfrak{h})$ consists of nilpotent operators as well, so we may apply the induction hypothesis to obtain a non-zero $\overline{a} \in \mathfrak{g}/\mathfrak{h}$ such that $\pi(\mathfrak{h})\overline{a} = 0$. Take any lift $a \in \mathfrak{g}$ of $\overline{a}$, and note that $[\mathfrak{h}, a] \subset \mathfrak{h}$. Since $\overline{a} \neq 0$, we have $a \notin \mathfrak{h}$, so $\mathfrak{h} \oplus \langle a \rangle$ is a larger subalgebra than $\mathfrak{h}$, and by maximality of $\mathfrak{h}$ we must have $\mathfrak{g} = \mathfrak{h} \oplus \langle a \rangle$. In particular, $\mathfrak{h}$ is an ideal in $\mathfrak{g}$ of codimension 1.

By the induction hypothesis, there exists some non-zero $v \in V$ such that $\mathfrak{h}v = 0$, so $W = \{v \in V : \mathfrak{h}v = 0\}$ is non-empty. Moreover, $W$ is invariant under $\mathfrak{g}$ since $hgw = [h, g]w + ghw = 0 + 0 = 0$ for all $w \in W$, $h \in \mathfrak{h}$ and $g \in \mathfrak{g}$. Using that $a$ is nilpotent, there exists some non-zero $w \in W$ such that $aw = 0$. Now $\mathfrak{h}w = 0$ and $aw = 0$ imply $\mathfrak{g}w = 0$ as desired. □

**Corollary 3.5.** *Let $\pi : \mathfrak{g} \to \mathfrak{gl}(V)$ be a finite-dimensional representation of a Lie algebra $\mathfrak{g}$, such that $\pi(\mathfrak{g})$ consists of nilpotent operators. Then there exists a basis for $V$ in which all operators in $\pi(\mathfrak{g})$ are strictly upper triangular.*

*Proof.* By induction on $n = \dim V$. By Engel's theorem, there exists a non-zero vector $e_1 \in V$ such that $\pi(\mathfrak{g})e_1 = 0$. In particular, $\langle e_1 \rangle$ is an invariant subspace, so we may apply the induction

hypothesis to the induced representation $\overline{\pi} : \mathfrak{g} \to \mathfrak{gl}(V/\langle e_1 \rangle)$ to obtain a basis $\overline{e}_2, \ldots, \overline{e}_n$ of $V/\langle e_1 \rangle$ in which all operators $\overline{\pi}(\mathfrak{g})$ are strictly upper triangular. For any preimages $e_2, \ldots, e_n$ of $\overline{e}_2, \ldots, \overline{e}_n$, the basis $e_1, e_2, \ldots, e_n$ for $V$ suffices. □

**Corollary 3.6.** *A finite-dimensional Lie algebra $\mathfrak{g}$ is nilpotent if and only if $\mathrm{ad}_g$ is nilpotent for all $g \in \mathfrak{g}$.*

*Proof.* The 'only if' part is obvious, and if all $\mathrm{ad}_g$ are nilpotent, then Engel's theorem can be applied to the faithful representation $\mathrm{ad} : \mathfrak{g}/Z(\mathfrak{g}) \to \mathfrak{gl}(\mathfrak{g})$ to give a basis for $\mathfrak{g}$ in which $\mathfrak{g}/Z(\mathfrak{g})$ consists of strictly upper triangular matrices. Hence, $\mathfrak{g}/Z(\mathfrak{g})$ is nilpotent, and so is $\mathfrak{g}$. □

# 4 Weight decompositions

**Definition 4.1.** Let $V$ be a vector space over a field $k$. The **(generalized) eigenspace** of a linear operator $a \in \mathfrak{gl}(V)$ with eigenvalue $\lambda \in k$ is the subspace

$$V_\lambda^a = \{v \in V : (a - \lambda)^N v = 0 \text{ for some } N \geq 0\}.$$

**Remark 4.2.** From linear algebra we know that over $k = \mathbb{C}$, we have a decomposition

$$V = \bigoplus_{\lambda \in \mathbb{C}} V_\lambda^a.$$

In particular, if $\mathfrak{g}$ is a finite-dimensional complex Lie algebra and $\pi : \mathfrak{g} \to \mathfrak{gl}(V)$ a finite-dimensional representation, we obtain such a decomposition for any $a \in \mathfrak{g}$. As a special case, for the adjoint representation, we obtain

$$\mathfrak{g} = \bigoplus_{\alpha \in \mathbb{C}} \mathfrak{g}_\alpha^a.$$

**Theorem 4.3.** *Let $\mathfrak{g}$ be a Lie algebra, and $\pi : \mathfrak{g} \to \mathfrak{gl}(V)$ a representation. Then $\pi(\mathfrak{g}_\alpha^a) V_\lambda^a \subset V_{\lambda+\alpha}^a$.*

*Proof.* For any $v \in V_\lambda^a$ and $g \in \mathfrak{g}_\alpha^a$, we have

$$(\pi(a) - \lambda - \alpha)^N \pi(g) v = \sum_{j=0}^{N} \binom{N}{j} (\mathrm{ad}_{\pi(a)} - \alpha)^j \pi(g) (\pi(a) - \lambda)^{N-j} v.$$

In particular, for $N > \dim V_\lambda^a + \dim \mathfrak{g}_\alpha^a$ the right-hand side vanishes. □

**Definition 4.4.** Let $\mathfrak{g}$ be a Lie algebra and $\pi : \mathfrak{g} \to \mathfrak{gl}(V)$ a representation. The **(generalized) weight space** of $\mathfrak{g}$ corresponding to some $\lambda \in \mathfrak{g}^*$ is the subspace

$$V_\lambda^{\mathfrak{g}} = \{v \in V : (\pi(g) - \lambda(g))^N v = 0 \text{ for some } N, \text{ for all } g \in \mathfrak{g}\}.$$

When $V_\lambda^{\mathfrak{g}} \neq 0$ we say $\lambda$ is a **weight** for $\pi$.

**Remark 4.5.** When $g_1, \ldots, g_n$ is a basis for $\mathfrak{g}$, a weight $\lambda \in \mathfrak{g}^*$ is nothing more than a sequence $(\lambda_1, \ldots, \lambda_n) \in k^n$ such that $\pi(g_i) v = \lambda_i v$ for all $i$ for some non-zero vector $v \in V$.

**Example 4.6.** For any subalgebra $\mathfrak{h} \subset \mathfrak{g}$, the restriction of the adjoint representation $\mathrm{ad}|_{\mathfrak{h}} : \mathfrak{h} \to \mathfrak{gl}(\mathfrak{g})$ defines (generalized) weight spaces

$$\mathfrak{g}_\lambda^{\mathfrak{h}} = \{x \in \mathfrak{g} : (\mathrm{ad}_h - \lambda(h))^N x = 0 \text{ for some } N, \text{ for all } h \in \mathfrak{h}\}$$

with $\lambda \in \mathfrak{h}^*$.

**Lemma 4.7** (Lie's lemma)**.** *Let $\mathfrak{g}$ be a complex Lie algebra, $\mathfrak{h} \subset \mathfrak{g}$ an ideal, and $\pi : \mathfrak{g} \to \mathfrak{gl}(V)$ a finite-dimensional representation. Then each weight space $V_\lambda^{\mathfrak{h}}$ for $\pi|_{\mathfrak{h}}$ is invariant under $\mathfrak{g}$.*

*Proof.* We must show for any $v \in V_\lambda^{\mathfrak{h}}$, $a \in \mathfrak{g}$ and $h \in \mathfrak{h}$ that $\pi(h)\pi(a)v = \lambda(h)\pi(a)v$. Since

$$\pi(h)\pi(a)v = \pi([h,a])v + \pi(a)\pi(h)v = \lambda([h,a])v + \lambda(h)\pi(a)v,$$

it suffices to show that $\lambda([h,a]) = 0$ whenever $V_\lambda^{\mathfrak{h}} \neq 0$. Consider the sequence of subspaces

$$\{0\} = W_{-1} \subset W_0 \subset W_1 \subset \cdots \qquad \text{with } W_m = \langle v, \pi(a)v, \ldots, \pi(a)^m v\rangle.$$

Since $V$ is finite-dimensional, there exists some minimal $N \geq 0$ for which $W_N = W_{N+1}$. By induction on $m$, we will show that $W_m$ is invariant under $\mathfrak{h}$, and moreover

$$\pi(h)\pi(a)^m v - \lambda(h)\pi(a)^m v \in W_{m-1} \qquad \text{for all } h \in \mathfrak{h}.$$

The case $m = 0$ is trivial, and for $m > 0$ we find

$$\pi(h)\pi(a)^m v - \lambda(h)\pi(a)^m v = [\pi(h), \pi(a)]\pi(a)^{m-1}v + \pi(a)\left(\pi(h)\pi(a)^{m-1}v - \lambda(h)\pi(a)^{m-1}v\right) \in W_{m-1}$$

because of $[\pi(h), \pi(a)] \in \pi(\mathfrak{h})$ and the induction hypothesis.

Now, $W_N$ is invariant under $\pi(a)$ and $\pi(\mathfrak{h})$, and in terms of the basis $\{v, \pi(a)v, \ldots, \pi(a)^N v\}$, any $\pi(h)$ with $h \in \mathfrak{h}$ acts on $W_N$ by an upper triangular matrix with $\lambda(h)$ all over the diagonal. Hence,

$$0 = \mathrm{tr}_{W_N}([\pi(h), \pi(a)]) = \mathrm{tr}_{W_N}(\pi([h,a])) = (N+1)\lambda([h,a]),$$

so $\lambda([h,a]) = 0$, which finishes the proof. □

**Theorem 4.8** (Lie's theorem)**.** *Let $\mathfrak{g}$ be a solvable complex Lie algebra and $\pi : \mathfrak{g} \to \mathfrak{gl}(V)$ a finite-dimensional representation with $V \neq 0$. Then there exists a weight $\lambda \in \mathfrak{g}^*$ for $\pi$.*

*Proof.* We may assume $\mathfrak{g}$ is finite-dimensional since $\pi$ factors through $\pi(\mathfrak{g})$, which is finite-dimensional and also solvable. Proof by induction on $\dim \mathfrak{g}$, where the case $\dim \mathfrak{g} = 0$ is trivial, so we may assume $\dim \mathfrak{g} > 0$. As $\mathfrak{g}$ is solvable, $[\mathfrak{g}, \mathfrak{g}] \subsetneq \mathfrak{g}$, so the quotient $\mathfrak{g}/[\mathfrak{g}, \mathfrak{g}]$ is abelian of dimension $\geq 1$. Let $\overline{\mathfrak{h}}$ be a subspace of this quotient of codimension one, then it is automatically an ideal and its preimage $\mathfrak{h} \subset \mathfrak{g}$ is also an ideal of codimension one. Thus, we can write $\mathfrak{g} = \mathfrak{h} \oplus \langle a \rangle$ for some $a \in \mathfrak{g}$. Now, $\mathfrak{h}$ is also solvable, so by the induction hypothesis there exists some weight $\lambda \in \mathfrak{h}^*$ for the restriction of $\pi$ to $\mathfrak{h}$. By Lie's lemma, the weight space $V_\lambda^{\mathfrak{h}}$ is invariant under $\pi(\mathfrak{g})$, so in particular $\pi(a)V_\lambda^{\mathfrak{h}} \subset V_\lambda^{\mathfrak{h}}$. As $\mathbb{C}$ is algebraically closed, there exists some non-zero $v \in V_\lambda^{\mathfrak{h}}$ such that $\pi(a)v = \ell a$ for some $\ell \in \mathbb{C}$. Now define $\lambda' \in \mathfrak{g}^*$ by $\lambda'(h + \mu a) = \lambda(h) + \mu\ell$ for all $h \in \mathfrak{h}$ and $\mu \in \mathbb{C}$. Then by construction $v \in V_{\lambda'}^{\mathfrak{g}}$, so $\lambda'$ is a weight for $\pi$. □

**Theorem 4.9.** *Let $\mathfrak{g}$ be a complex finite-dimensional Lie algebra, $\pi : \mathfrak{g} \to \mathfrak{gl}(V)$ a representation, and $\mathfrak{h}$ a nilpotent subalgebra. Then $V$ decomposes as*

$$V = \bigoplus_{\lambda \in \mathfrak{h}^*} V_\lambda^{\mathfrak{h}}, \qquad \textit{with} \qquad \pi\left(\mathfrak{g}_\alpha^{\mathfrak{h}}\right) V_\lambda^{\mathfrak{h}} \subset V_{\lambda+\alpha}^{\mathfrak{h}}.$$

*In particular, for the adjoint representation,*

$$\mathfrak{g} = \bigoplus_{\alpha \in \mathfrak{h}^*} \mathfrak{g}_\alpha^{\mathfrak{h}} \qquad \textit{with} \qquad [\mathfrak{g}_\alpha^{\mathfrak{h}}, \mathfrak{g}_\beta^{\mathfrak{h}}] \subset \mathfrak{g}_{\alpha+\beta}^{\mathfrak{h}}.$$

*Proof.* By induction on $\dim V$, where the case $V = 0$ is trivial, so we may assume that $V \neq 0$. First consider the case where for all $h \in \mathfrak{h}$, the operator $\pi(h)$ has only one eigenvalue, denoted $\lambda(h)$. Then we must have that $V = V_{\lambda(h)}^h$ for all $h \in \mathfrak{h}$, and it remains to check linearity of $\lambda$. Since $\mathfrak{h}$ is nilpotent, it is solvable, so by Lie's theorem (4.8) there exists at least one weight $\lambda' \in \mathfrak{h}^*$ with non-zero weight space $V_{\lambda'}^{\mathfrak{h}}$. By assumption, $\lambda'(h)$ must be the (only) eigenvalue of $\pi(h)$, and hence $\lambda = \lambda'$ is linear.

Now consider the case that $\pi(a)$ has at least two distinct eigenvalues for some $a \in \mathfrak{h}$, and decompose $V = \bigoplus_\lambda V_\lambda^a$. As $\mathfrak{h} \subset \mathfrak{g}_0^a$, we have $\pi(\mathfrak{h}) V_\lambda^a \subset V_\lambda^a$, so each $V_\lambda^a$ is also a representation of $\mathfrak{h}$. Since $\pi(a)$ has multiple eigenvalues, we have $\dim V_\lambda^a < \dim V$, and we may proceed by induction.

Finally, the inclusions $\pi\left(\mathfrak{g}_\alpha^{\mathfrak{h}}\right) V_\lambda^{\mathfrak{h}} \subset V_{\lambda+\alpha}^{\mathfrak{h}}$ follow from Theorem 4.3. □

# 5 Cartan subalgebras

**Definition 5.1.** Let $\mathfrak{h}$ be a subalgebra of $\mathfrak{g}$. Then $N_{\mathfrak{g}}(\mathfrak{h}) = \{a \in \mathfrak{g} : [a, \mathfrak{h}] \subset \mathfrak{h}\}$ is also a subalgebra of $\mathfrak{g}$, called the **normalizer** of $\mathfrak{h}$.

**Lemma 5.2.** *For any nilpotent Lie algebra $\mathfrak{g}$ and proper subalgebra $\mathfrak{h} \subsetneq \mathfrak{g}$ we have $\mathfrak{h} \subsetneq N_{\mathfrak{g}}(\mathfrak{h})$.*

*Proof.* Since the lower central series $\mathfrak{g}^1 \supset \mathfrak{g}^2 \supset \cdots \supset \mathfrak{g}^n = 0$ terminates for some $n$, let $j$ be the maximal integer such that $\mathfrak{g}^j \not\subset \mathfrak{h}$. Then $[\mathfrak{g}^j, \mathfrak{h}] \subset \mathfrak{g}^{j+1} \subset \mathfrak{h}$, so $\mathfrak{g}^j \subset N_{\mathfrak{g}}(\mathfrak{h})$. We conclude that $\mathfrak{h} \subsetneq N_{\mathfrak{g}}(\mathfrak{h})$. □

**Definition 5.3.** A **Cartan subalgebra** of a Lie algebra $\mathfrak{g}$ is a nilpotent subalgebra $\mathfrak{h}$ with $N_{\mathfrak{g}}(\mathfrak{h}) = \mathfrak{h}$.

In particular, by Lemma 5.2 any Cartan subalgebra is a maximal nilpotent subalgebra. Also, a Lie algebra $\mathfrak{g}$ is a Cartan subalgebra of itself if and only if $\mathfrak{g}$ is nilpotent.

**Example 5.4.** Let $\mathfrak{g} \subset \mathfrak{gl}_n(k)$ be a subalgebra with containing a diagonal matrix $a = \operatorname{diag}(a_1, \ldots, a_n)$ with distinct $a_i$. Then the subalgebra $\mathfrak{h} \subset \mathfrak{g}$ consisting of all diagonal matrices is a Cartan subalgebra. Indeed, since $\mathfrak{h}$ is abelian it is nilpotent. Now let $b = \sum_{i,j} b_{ij} e_{ij} \in \mathfrak{g}$ be such that $[b, \mathfrak{h}] \subset \mathfrak{h}$, that is $b \in N_{\mathfrak{g}}(\mathfrak{h})$. In particular $[b, a] = \sum_{ij} (a_j - a_i) b_{ij} e_{ij} \in \mathfrak{h}$, which is non-diagonal only if $b_{ij} \neq 0$ for some $i \neq j$. Hence $b \in \mathfrak{h}$ and thus $N_{\mathfrak{g}}(\mathfrak{h}) = \mathfrak{h}$.

**Example 5.5.** Let $\mathfrak{g} = \mathfrak{su}(2), \mathfrak{gl}_n, \mathfrak{sl}_n$. Cartan algebra = ?

**Proposition 5.6.** *Let $\mathfrak{g}$ be a finite-dimensional complex Lie algebra, $\mathfrak{h} \subset \mathfrak{g}$ a Cartan subalgebra, and consider the decomposition $\mathfrak{g} = \bigoplus_{\alpha \in \mathfrak{h}^*} \mathfrak{g}_\alpha$. Then $\mathfrak{g}_0 = \mathfrak{h}$.*

*Proof.* Since $\mathfrak{h}$ is nilpotent, Corollary 3.6 implies that all $\mathrm{ad}_h|_{\mathfrak{h}}$ are nilpotent for $h \in \mathfrak{h}$, so $\mathfrak{h} \subset \mathfrak{g}_0$. If $h \neq \mathfrak{g}_0$, then applying Engel's theorem to $\mathrm{ad} : \mathfrak{h} \to \mathfrak{gl}(\mathfrak{g}_0/\mathfrak{h})$ yields a non-zero $\overline{a} \in \mathfrak{g}_0/\mathfrak{h}$, which lifts to some $a \in \mathfrak{g}_0$ with $[a, \mathfrak{h}] \subset \mathfrak{h}$. But then $\mathfrak{h} \neq N_{\mathfrak{g}}(\mathfrak{h})$, contradicting that $\mathfrak{h}$ is a Cartan subalgebra. $\square$

**Remark 5.7.** By the above proposition, a finite-dimensional complex Lie algebra decomposes as

$$\mathfrak{g} = \mathfrak{h} \oplus \bigoplus_{\alpha \in \mathfrak{h}^* \setminus \{0\}} \mathfrak{g}_\alpha.$$

# 6 Cartan's criterion

**Definition 6.1.** Let $\pi : \mathfrak{g} \to \mathfrak{gl}(V)$ be a finite-dimensional representation of a Lie algebra $\mathfrak{g}$. The associated **trace form** is a bilinear form on $\mathfrak{g}$ given by

$$(x, y)_V = \mathrm{tr}(\pi(x)\pi(y)).$$

**Definition 6.2.** The **Killing form** of a finite-dimensional Lie algebra $\mathfrak{g}$, is the trace form of its adjoint representation, that is,

$$\kappa_{\mathfrak{g}}(x, y) = \mathrm{tr}(\mathrm{ad}_x \circ \mathrm{ad}_y)$$

**Proposition 6.3.** *The trace form $(\cdot, \cdot)_V$ is symmetric and associative, that is,*

$$(x, y)_V = (y, x)_V \quad \text{and} \quad ([x, y], z])_V = (x, [y, z])_V$$

*for all $x, y, z \in \mathfrak{g}$.*

*Proof.* Symmetry follows from the cyclic property of the trace, and associativity follows from

$$\begin{aligned}([x, y], z)_V &= \mathrm{tr}(\pi([x, y])\pi(z)) = \mathrm{tr}(\pi(x)\pi(y)\pi(z) - \pi(y)\pi(x)\pi(z)) \\ &= \mathrm{tr}(\pi(x)\pi(y)\pi(z) - \pi(x)\pi(z)\pi(y)) = \mathrm{tr}(\pi(x)\pi([y, z])) = (x, [y, z])_V.\end{aligned}$$

$\square$

**Lemma 6.4** (Cartan)**.** *Let $\mathfrak{g}$ be a finite-dimensional complex Lie algebra, and $\pi : \mathfrak{g} \to \mathfrak{gl}(V)$ a finite-dimensional representation. Let $\mathfrak{h} \subset \mathfrak{g}$ be a Cartan subalgebra, and consider the generalized weight space decompositions of $V$ and $\mathfrak{g}$,*

$$V = \bigoplus_{\lambda \in \mathfrak{h}^*} V_\lambda^{\mathfrak{h}}, \quad \mathfrak{g} = \bigoplus_{\alpha \in \mathfrak{h}^*} \mathfrak{g}_\alpha^{\mathfrak{h}}.$$

*Take any $\alpha \in \mathfrak{h}^*$ and $e \in \mathfrak{g}_\alpha^{\mathfrak{h}}$ and $f \in \mathfrak{g}_{-\alpha}^{\mathfrak{h}}$ with $h = [e, f] \in \mathfrak{g}_0^{\mathfrak{h}} = \mathfrak{h}$. If $V_\lambda^{\mathfrak{h}} \neq 0$ then $\lambda(h) = r\alpha(h)$ for some $r \in \mathbb{Q}$ only depending on $\lambda$ and $\alpha$.*

*Proof.* Let $U = \bigoplus_{n\in\mathbb{Z}} V^{\mathfrak{h}}_{\lambda+n\alpha} \subset V$. Then $U$ is finite-dimensional, and invariant under $\pi(e), \pi(f)$ and $\pi(h)$. Since $[\pi(e), \pi(f)] = \pi(h)$, we have

$$0 = \mathrm{tr}_U(\pi(h)) = \sum_{n\in\mathbb{Z}} \mathrm{tr}_{V^{\mathfrak{h}}_{\lambda+n\alpha}}(\pi(h)) = \sum_{n\in\mathbb{Z}} (\lambda + n\alpha)(h) \dim V^{\mathfrak{h}}_{\lambda+n\alpha},$$

and thus

$$\lambda(h) = r\alpha(h), \qquad \text{with} \qquad r = -\frac{\sum_{n\in\mathbb{Z}} n \ \dim V^{\mathfrak{h}}_{\lambda+n\alpha}}{\sum_{n\in\mathbb{Z}} \dim V^{\mathfrak{h}}_{\lambda+n\alpha}}.$$

□

**Theorem 6.5** (Cartan's criterion)**.** *Let $\mathfrak{g} \subset \mathfrak{gl}(V)$ for some finite-dimensional complex vector space $V$. Then the following are equivalent.*

*(i)* $(x, y)_V = 0$ *for all* $x \in \mathfrak{g}$ *and* $y \in [\mathfrak{g}, \mathfrak{g}]$.

*(ii)* $(x, x)_V = 0$ *for all* $x \in [\mathfrak{g}, \mathfrak{g}]$.

*(iii)* $\mathfrak{g}$ *is solvable.*

*Proof.* ($i \Rightarrow ii$) Obvious. ($iii \Rightarrow i$) After a base change to an algebraically closed field, one can apply Lie's theorem (4.8) to obtain a basis of $V$ in which $\mathfrak{g}$ consists of upper triangular matrices, and hence $[\mathfrak{g}, \mathfrak{g}]$ of strictly upper triangular matrices. Then it is clear that $(x, y)_V = \mathrm{tr}(xy) = 0$ for all $x \in \mathfrak{g}$ and $y \in [\mathfrak{g}, \mathfrak{g}]$, since $xy$ is also strictly upper triangular. ($ii \Rightarrow iii$) If $\mathfrak{g}$ is not solvable, then the derived series stabilizes, that is, $[\mathfrak{g}^{(i)}, \mathfrak{g}^{(i)}] = \mathfrak{g}^{(i)} \neq 0$ for some $i$. Hence, $(x, x)_V = 0$ for all $x \in \mathfrak{g}^{(i)}$, and we reach a contradiction using the following lemma. □

**Lemma 6.6.** *Let $\mathfrak{g} \subset \mathfrak{gl}(V)$ for some finite-dimensional complex vector space $V$, such that $[\mathfrak{g}, \mathfrak{g}] = \mathfrak{g}$. Then $(x, x)_V \neq 0$ for some $x \in \mathfrak{g}$.*

*Proof.* Assume for a contradiction that $(x, x)_V = 0$ for all $x \in \mathfrak{g}$. For any $\lambda \in \mathfrak{h}^*$, write $\lambda(h) = r_{\lambda,\alpha}\alpha(h)$ with $r_{\lambda,\alpha} \in \mathbb{Q}$ as in Cartan's lemma (6.4). Now for any $h \in [\mathfrak{g}_\alpha, \mathfrak{g}_{-\alpha}]$ we have

$$0 = (h, h)_V = \mathrm{tr}_V(\pi(h)^2) = \sum_{\lambda\in\mathfrak{h}^*} \mathrm{tr}_{V_\lambda}(\pi(h)^2) = \sum_{\lambda\in\mathfrak{h}^*} \lambda(h)^2 \dim V_\lambda = \alpha(h)^2 \sum_{\lambda\in\mathfrak{h}^*} r^2_{\lambda,\alpha} \dim V_\lambda,$$

so in particular $\alpha(h) = 0$ and $\lambda(h) = 0$. From $[\mathfrak{g}, \mathfrak{g}] = \mathfrak{g}$ and $[\mathfrak{g}_\alpha, \mathfrak{g}_\beta] \subset \mathfrak{g}_{\alpha+\beta}$ we have $\mathfrak{h} = \sum_{\alpha\in\mathfrak{h}^*} [\mathfrak{g}_\alpha, \mathfrak{g}_{-\alpha}]$, so the above implies that $\lambda(h) = 0$ for all $h \in \mathfrak{h}$, i.e. $\lambda = 0$. Since $\lambda$ was arbitrary, we have $V = V_0$. Finally, $\mathfrak{g}_\alpha V = \mathfrak{g}_\alpha V_0 \subset V_\alpha$ shows that $\mathfrak{g}_\alpha = 0$ for $\alpha \neq 0$, so that $\mathfrak{g} = \mathfrak{g}_0 = \mathfrak{h}$ is nilpotent, contradicting the assumption that $[\mathfrak{g}, \mathfrak{g}] = \mathfrak{g}$. □

**Corollary 6.7.** *A finite-dimensional complex Lie algebra $\mathfrak{g}$ is solvable if and only if $\kappa_{\mathfrak{g}}(\mathfrak{g}, [\mathfrak{g}, \mathfrak{g}]) = 0$.*

*Proof.* Apply the above proposition to the representation $\mathrm{ad} : \mathfrak{g}/Z(\mathfrak{g}) \to \mathfrak{gl}(\mathfrak{g})$. □

# 7 Semisimple Lie algebras

**Definition 7.1.** A **radical** $\mathrm{Rad}(\mathfrak{g})$ of a finite-dimensional Lie algebra $\mathfrak{g}$ is a solvable ideal of $\mathfrak{g}$ of maximal possible dimension.

**Proposition 7.2.** *There exists a unique radical* $\mathrm{Rad}(\mathfrak{g})$ *of* $\mathfrak{g}$*, and it contains all solvable ideals of* $\mathfrak{g}$*.*

*Proof.* If $\mathfrak{a} \subset \mathfrak{g}$ is a solvable ideal, then $\mathfrak{a} + \mathrm{Rad}(\mathfrak{g})$ is again a solvable, so $\mathfrak{a} = \mathrm{Rad}(\mathfrak{g})$ by maximality. This also shows that any two radicals would contain each other, and thus be equal. □

**Remark 7.3.** If $\mathfrak{g}$ is finite-dimensional solvable Lie algebra, then $\mathrm{Rad}(\mathfrak{g}) = \mathfrak{g}$.

**Definition 7.4.** A finite-dimensional Lie algebra $\mathfrak{g}$ is **semisimple** if $\mathrm{Rad}(\mathfrak{g}) = 0$.

**Proposition 7.5.** *Let* $\mathfrak{g}$ *be a finite-dimensional Lie algebra. Then the following are equivalent.*

*(i)* $\mathfrak{g}$ *is semisimple,*

*(ii) any solvable ideal of* $\mathfrak{g}$ *is zero,*

*(iii) any abelian ideal of* $\mathfrak{g}$ *is zero.*

*Proof.* $(i \Rightarrow ii)$ Any solvable ideal $\mathfrak{a} \subset \mathfrak{g}$ is contained in $\mathrm{Rad}(\mathfrak{g}) = 0$, so $\mathfrak{a} = 0$. $(ii \Rightarrow iii)$ Any abelian ideal is solvable. $(iii \Rightarrow i)$ Suppose that $\mathfrak{r} = \mathrm{Rad}(\mathfrak{g})$ is non-zero. Since the radical is solvable, we have

$$\mathfrak{r} \supset \mathfrak{r}^{(1)} \supset \mathfrak{r}^{(2)} \supset \cdots \supsetneq \mathfrak{r}^{(k)} = 0$$

for some $k$. In particular, $\mathfrak{r}^{(k-1)}$ is a non-zero abelian ideal. □

**Remark 7.6.** Let $\mathfrak{g}$ be a finite-dimensional Lie algebra. The quotient $\mathfrak{f} = \mathfrak{g}/\mathrm{Rad}(\mathfrak{g})$ is a semisimple Lie algebra. In this sense, being semisimple is opposite to being solvable.

$$0 \to \mathrm{Rad}(\mathfrak{g}) \to \mathfrak{g} \to \mathfrak{g}/\mathrm{Rad}(\mathfrak{g}) \to 0$$

In characteristic zero, a stronger result holds.

**Theorem 7.7** (Levi decomposition)**.** *Let* $\mathfrak{g}$ *be a finite-dimensional Lie algebra over a field of characteristic zero. Then there exists a semisimple subalgebra* $\mathfrak{s} \subset \mathfrak{g}$ *such that* $\mathfrak{g} = \mathfrak{s} \oplus \mathrm{Rad}(\mathfrak{g})$ *as vector spaces.*

**Theorem 7.8.** *A finite-dimensional Lie algebra* $\mathfrak{g}$ *over a field of characteristic zero is semisimple if and only if the Killing form* $\kappa_{\mathfrak{g}}$ *is non-degenerate.*

*Proof.* $(\Leftarrow)$ Let $\mathfrak{a} \subset \mathfrak{g}$ be an abelian ideal. For any $x \in \mathfrak{a}$ and $y \in \mathfrak{g}$, we have $\mathrm{ad}_x \mathrm{ad}_y = [x, [y, -]]$ maps to $\mathfrak{a}$, and $(\mathrm{ad}_x \mathrm{ad}_y)^2 = 0$ since $[\mathfrak{a}, \mathfrak{a}] = 0$. Hence, $\mathrm{tr}(\mathrm{ad}_x \mathrm{ad}_y) = \kappa_{\mathfrak{g}}(x, y) = 0$, so $x = 0$ by non-degeneracy of $\kappa_{\mathfrak{g}}$, implying $\mathfrak{a} = 0$. Hence $\mathfrak{g}$ is semisimple.

($\Rightarrow$) Suppose that $\mathfrak{g}$ is semisimple. Note that $\mathfrak{s} = \{x \in \mathfrak{g} : \kappa_{\mathfrak{g}}(x, -) = 0\}$ is an ideal of $\mathfrak{g}$ since

$$\kappa_{\mathfrak{g}}([g, x], -) = -\kappa_{\mathfrak{g}}([x, g], -) = -\kappa_{\mathfrak{g}}(x, [g, -]) = 0$$

for any $x \in \mathfrak{s}$ and $g \in \mathfrak{g}$. Obviously, $\kappa_{\mathfrak{s}}(x, y) = \kappa_{\mathfrak{g}}(x, y) = 0$ for any $x \in \mathfrak{s}$ and $y \in [\mathfrak{s}, \mathfrak{s}]$, so Cartan's criterion (6.5) yields that $\mathfrak{s}$ is solvable, and thus zero. $\square$

**Definition 7.9.** A Lie algebra $\mathfrak{g}$ is **simple** if it is non-abelian and its only ideals are 0 and $\mathfrak{g}$.

**Theorem 7.10.** *Any semisimple finite-dimensional Lie algebra $\mathfrak{g}$ over a field of characteristic zero is a direct sum of its simple ideals.*

*Proof.* For any ideal $\mathfrak{a} \subset \mathfrak{g}$, the orthogonal complement $\mathfrak{a}^{\perp} = \{x \in \mathfrak{g} : \kappa_{\mathfrak{g}}(x, \mathfrak{a}) = 0\}$ is also an ideal of $\mathfrak{g}$. Indeed, $\kappa_{\mathfrak{g}}([g, x], a) = -\kappa_{\mathfrak{g}}(x, [g, a]) = 0$ for all $x \in \mathfrak{a}^{\perp}$, $g \in \mathfrak{g}$ and $a \in \mathfrak{a}$. Moreover, for any $x \in \mathfrak{a} \cap \mathfrak{a}^{\perp}$ and $y \in [\mathfrak{a} \cap \mathfrak{a}^{\perp}, \mathfrak{a} \cap \mathfrak{a}^{\perp}]$ we have $0 = \kappa_{\mathfrak{g}}(x, y) = \kappa_{\mathfrak{a} \cap \mathfrak{a}^{\perp}}(x, y)$, so by Cartan's criterion (6.5), $\mathfrak{a} \cap \mathfrak{a}^{\perp}$ is solvable and hence zero. As $\kappa_{\mathfrak{g}}$ is non-degenerate, we have a decomposition $\mathfrak{g} = \mathfrak{a} \oplus \mathfrak{a}^{\perp}$ as vector spaces, which is automatically a direct sum of Lie algebras since $[\mathfrak{a}, \mathfrak{a}^{\perp}] \subset \mathfrak{a} \cap \mathfrak{a}^{\perp} = 0$.

By induction, we can repeat this argument until we obtain a decomposition $\mathfrak{g} = \mathfrak{a}_1 \oplus \cdots \oplus \mathfrak{a}_{\ell}$ in which all summands are simple ideals. To see why the $\mathfrak{a}_i$ are the only simple ideals of $\mathfrak{g}$, let $\mathfrak{b} \subset \mathfrak{g}$ be any simple ideal, and note that

$$\mathfrak{b} = [\mathfrak{b}, \mathfrak{g}] = [\mathfrak{b}, \mathfrak{a}_1 \oplus \cdots \oplus \mathfrak{a}_{\ell}] = [\mathfrak{b}, \mathfrak{a}_1] \oplus \cdots \oplus [\mathfrak{b}, \mathfrak{a}_{\ell}].$$

Finally, as $\mathfrak{b}$ is simple, we must have $\mathfrak{b} = [\mathfrak{b}, \mathfrak{g}] \subset \mathfrak{a}_i$ for some $i$, and since $\mathfrak{a}_i$ is simple, $\mathfrak{b} = \mathfrak{a}_i$. $\square$

# 8 Root systems of semisimple Lie algebras

**Theorem 8.1.** *Let $\mathfrak{g}$ be a finite-dimensional semisimple Lie algebra, and $\mathfrak{h}$ a Cartan subalgebra. Consider the decomposition $\mathfrak{g} = \bigoplus_{\alpha \in \mathfrak{h}^*} \mathfrak{g}_{\alpha}$ with $\mathfrak{g}_0 = \mathfrak{h}$. Then*

*(i)* $\kappa_{\mathfrak{g}}(\mathfrak{g}_{\alpha}, \mathfrak{g}_{\beta}) = 0$ *for* $\alpha + \beta \neq 0$,

*(ii)* $\kappa_{\mathfrak{g}}|_{\mathfrak{g}_{\alpha} \times \mathfrak{g}_{-\alpha}}$ *is non-degenerate,*

*(iii)* $\kappa_{\mathfrak{g}}|_{\mathfrak{h} \times \mathfrak{h}}$ *is non-degenerate,*

*(iv)* $\mathfrak{h}$ *is an abelian subalgebra of* $\mathfrak{g}$,

*(v)* $\mathrm{ad}_h$ *is diagonalizable for all* $h \in \mathfrak{h}$. *Consequently,* $\mathfrak{g}_{\alpha} = \{x \in \mathfrak{g} : [h, x] = \alpha(h)x \text{ for all } h \in \mathfrak{h}\}$.

*Proof.* (i) For any $x \in \mathfrak{g}_{\alpha}$ and $y \in \mathfrak{g}_{\beta}$ with $\alpha + \beta \neq 0$ we have

$$(\mathrm{ad}_x \mathrm{ad}_y)^N \mathfrak{g}_{\gamma} \subset \mathfrak{g}_{\gamma + N(\alpha + \beta)} = 0$$

for $N$ sufficiently large, as $\mathfrak{g}$ is finite-dimensional. Hence, $\mathrm{ad}_x \mathrm{ad}_y$ is nilpotent, and thus $\kappa_{\mathfrak{g}}(x, y) = \mathrm{tr}(\mathrm{ad}_x \mathrm{ad}_y) = 0$.

(ii) By semisimplicity, $\kappa_{\mathfrak{g}}$ is non-degenerate on $\mathfrak{g}$. From (i) we have $\kappa_{\mathfrak{g}}(\mathfrak{g}_\alpha, \mathfrak{g}_\beta) = 0$ when $\alpha \neq -\beta$, so $\kappa_{\mathfrak{g}}|_{\mathfrak{g}_\alpha \times \mathfrak{g}_{-\alpha}}$ must necessarily be non-degenerate.

(iii) Follows from (ii) as $\mathfrak{h} = \mathfrak{g}_0$.

(iv) Since $\mathfrak{h}$ is nilpotent, it is solvable, so Cartan's criterion ($\mathfrak{h} \subset \mathfrak{gl}(\mathfrak{g})$) yields

$$\kappa_{\mathfrak{g}}(x, y) = \mathrm{tr}(\mathrm{ad}_x \mathrm{ad}_y) = 0$$

for all $x \in \mathfrak{h}$ and $y \in [\mathfrak{h}, \mathfrak{h}]$. As $\kappa_{\mathfrak{g}}|_{\mathfrak{h}\times\mathfrak{h}}$ is non-degenerate by (iii), it follows that $[\mathfrak{h}, \mathfrak{h}] = 0$.

(v) By semisimplicity, the center $Z(\mathfrak{g})$ is zero (as it is an abelian ideal), so we can view $\mathfrak{g} \subset \mathfrak{gl}(\mathfrak{g})$ via the adjoint representation. Moreover, since $\mathfrak{h}$ is solvable (even nilpotent), there exists by Lie's theorem a basis for $\mathfrak{gl}(\mathfrak{g})$ such that $\mathfrak{h}$ consists of upper triangular matrices. Now decompose any $h \in \mathfrak{h}$ as $h = d + n$ with $d$ diagonal and $n$ strictly upper triangular. Since $[h, \mathfrak{h}] = 0$ also $[d, \mathfrak{h}] = 0$ so $d \in \mathfrak{h}$ by maximality of the Cartan subalgebra $\mathfrak{h}$. Since $n$ is strictly upper triangular, $\kappa_{\mathfrak{g}}(\mathfrak{h}, n) = 0$, so by (iii) we have $n = 0$, and thus $h = d$ is diagonalizable.

□

**Definition 8.2.** Let $\mathfrak{g}$ be a finite-dimensional semisimple Lie algebra, and $\mathfrak{h} \subset \mathfrak{g}$ a Cartan subalgebra. We can write

$$\mathfrak{g} = \mathfrak{h} \oplus \bigoplus_{\alpha \in \Phi} \mathfrak{g}_\alpha,$$

where $\Phi = \{\alpha \in \mathfrak{h}^* : \alpha \neq 0 \text{ and } \mathfrak{g}_\alpha \neq 0\}$ is called the **root system**. Elements $\alpha \in \Phi$ are called **roots**. In other words, a root is a non-zero weight of the adjoint representation $\mathrm{ad} : \mathfrak{g} \to \mathfrak{gl}(\mathfrak{g})$.

**Definition 8.3.** The restricted Killing form $\kappa_{\mathfrak{g}}|_{\mathfrak{h}\times\mathfrak{h}}$ defines an isomorphism $\nu : \mathfrak{h} \to \mathfrak{h}^*$ via $h \mapsto \kappa_{\mathfrak{g}}(h, \cdot)$. Write $t_\alpha = \nu^{-1}(\alpha)$, and define a bilinear form

$$(\cdot, \cdot) : \mathfrak{h}^* \times \mathfrak{h}^* \to k, \quad (\alpha, \beta) = \kappa_{\mathfrak{g}}(t_\alpha, t_\beta).$$

**Theorem 8.4.** *Let $\mathfrak{g}$ be a semisimple Lie algebra.*

*(i) If $\alpha \in \Phi$ then $-\alpha \in \Phi$ as well.*

*(ii) The subspace $[\mathfrak{g}_\alpha, \mathfrak{g}_{-\alpha}] \subset \mathfrak{g}_0$ is one-dimensional for all $\alpha \in \Phi$.*

*(iii) $(t_\alpha, t_\alpha) \neq 0$ for all $\alpha \in \Phi$.*

*(iv) For every $\alpha \in \Phi$ and $e \in \mathfrak{g}_\alpha$ there exists a $f \in \mathfrak{g}_{-\alpha}$ and $h \in \mathfrak{h}$ such that $\langle e, f, h\rangle$ is isomorphic to $\mathfrak{sl}_2(k)$.*

*Proof.* (i) If $-\alpha \notin \Phi$, then $g_\alpha$ would be orthogonal to the whole of $\mathfrak{g}$, contradicting the non-degeneracy of $\kappa_{\mathfrak{g}}$.

(ii) For any $x \in \mathfrak{g}_\alpha$ and $y \in \mathfrak{g}_{-\alpha}$ and $h \in \mathfrak{h}$,

$$\begin{aligned}\kappa_{\mathfrak{g}}(h, [x, y]) &= \kappa_{\mathfrak{g}}([h, x], y) \\ &= \alpha(h)\kappa_{\mathfrak{g}}(x, y) \\ &= \kappa_{\mathfrak{g}}(h, t_\alpha)\kappa_{\mathfrak{g}}(x, y) \\ &= \kappa_{\mathfrak{g}}(h, \kappa_{\mathfrak{g}}(x, y)t_\alpha),\end{aligned}$$

so it follows that $[x, y] = \kappa_{\mathfrak{g}}(x, y)t_\alpha$, and thus $[\mathfrak{g}^\alpha, \mathfrak{g}^{-\alpha}]$ is the one-dimensional subspaces spanned by $t_\alpha$. Note that $\kappa_{\mathfrak{g}}(x, y)$ cannot be zero for all $x, y$, as that would contradict the non-degeneracy of $\kappa_{\mathfrak{g}}$.

(iii) As $\kappa_{\mathfrak{g}}|_{\mathfrak{g}_\alpha \times \mathfrak{g}_{-\alpha}}$ is non-degenerate, we can find $e \in \mathfrak{g}_\alpha$ and $f \in \mathfrak{g}_{-\alpha}$ such that $\kappa_{\mathfrak{g}}(e, f) = 1$. From (ii) we know $[e, f] = t_\alpha$, and furthermore we compute

$$\begin{aligned}[t_\alpha, e] &= \alpha(t_\alpha)e = (\alpha, \alpha)e, \\ [t_\alpha, f] &= -\alpha(t_\alpha)f = -(\alpha, \alpha)f.\end{aligned}$$

If $(\alpha, \alpha) = 0$, then $\mathfrak{a} = \langle e, f, t_\alpha \rangle$ would be a solvable Lie algebra. Then by Lie's theorem, there exists a basis for $\mathfrak{gl}(\mathfrak{g})$ such that $\mathfrak{a}$ is upper triangular, and thus $t_\alpha \in [\mathfrak{a}, \mathfrak{a}]$ is strictly upper triangular. But we already know that $t_\alpha \in \mathfrak{h}$ is diagonalizable, so we must have $t_\alpha = 0$, implying $\alpha = 0$, a contradiction.

(iv) Pick $f$ such that $\kappa_{\mathfrak{g}}(e, f) = \frac{2}{(\alpha,\alpha)}$, and $h = \frac{2t_\alpha}{(\alpha,\alpha)}$. Similar to the above, we find

$$[e, f] = \kappa_{\mathfrak{g}}(e, f)t_\alpha = h, \quad [h, e] = \alpha(h)e = 2e, \quad [h, f] = -\alpha(h)f = -2f.$$

□

We can use the representation theory of $\mathfrak{sl}_2$ to make more conclusions about the root system.

**Theorem 8.5.** *Let $\mathfrak{g}$ be a semisimple complex Lie algebra, and $\mathfrak{h}$ a Cartan subalgebra.*

*(i)* $\dim \mathfrak{g}_\alpha = 1$ *for all* $\alpha \in \Phi$.

*(ii) If $\alpha, \beta \in \Phi$, then*

$$(\beta + \alpha\mathbb{Z}) \cap (\Phi \cup \{0\}) = \{\beta - p\alpha, \beta - (p-1)\alpha, \ldots, \beta + (q-1)\alpha, \beta + q\alpha\}$$

*for some $p, q \in \mathbb{Z}_{\geq 0}$ with $p - q = \frac{2(\alpha,\beta)}{(\alpha,\alpha)}$.*

*(iii) If $\alpha, \beta, \alpha + \beta \in \Phi$, then $[\mathfrak{g}_\alpha, \mathfrak{g}_\beta] = \mathfrak{g}_{\alpha+\beta}$.*

*(iv) If $\alpha \in \Phi$, then $n\alpha \in \Phi$ if and only if $n = \pm 1$.*

*Proof.* 1. Suppose that $\dim \mathfrak{g}_\alpha > 1$, then $\dim \mathfrak{g}_{-\alpha} = \dim \mathfrak{g}_\alpha > 1$ as well. Consider the subalgebra $\mathfrak{a}_\alpha = \langle e, f, h, \rangle$ of $\mathfrak{g}_\alpha$ isomorphic to $\mathfrak{sl}_2$, and its adjoint action on $\mathfrak{g}$. Since $\dim \mathfrak{g}_{-\alpha} > 1$, there exists some non-zero $v \in \mathfrak{g}_{-\alpha}$ such that $\kappa_{\mathfrak{g}}(e, v) = 0$, and hence $\mathrm{ad}_e(v) = [e, v] = \kappa_{\mathfrak{g}}(e, v)t_\alpha = 0$. However, $\mathrm{ad}_h(v) = [h, v] = \frac{2[t_\alpha, v]}{(\alpha,\alpha)} = \frac{-2(\alpha,\alpha)v}{(t_\alpha,t_\alpha)} = -2v$. From the finite-dimensional representation theory of $\mathfrak{sl}_2$ this causes a contradiction.

2. Let $q$ be the largest integer such that $\beta + q\alpha \in \Phi$. In particular, $q \geq 1$ as $\alpha + \beta \in \Phi$. Pick a non-zero vector $v \in \mathfrak{g}_{\beta+q\alpha}$ and note that $\mathrm{ad}_e(v) \in \mathfrak{g}_{\beta+(q+1)\alpha} = 0$. Also, $\mathrm{ad}_h(v) = (\beta + q\alpha)(h)v = \left(\frac{2(\alpha,\beta)}{(\alpha,\alpha)} + 2q\right) v$. By the finite-dimensional representation theory of $\mathfrak{sl}_2$ we find that $\lambda := \frac{2(\alpha,\beta)}{(\alpha,\alpha)} + 2q \in \mathbb{Z}_{\geq 0}$ and that $(\mathrm{ad}_f)^j(v)$ are non-zero vectors for $0 \leq j \leq \lambda$. Since $(\mathrm{ad}_f)^j(v) \in \mathfrak{g}_{\beta+(q-j)\alpha}$, we have $\beta + (q-j)\alpha \in \Phi$ for all $0 \leq j \leq \lambda$. Write $p = -(q - \lambda) = q + \frac{2(\alpha,\beta)}{(\alpha,\alpha)}$. Finish the proof.

3. Let $p$ and $q$ be as above, and pick a non-zero vector $v \in \mathfrak{g}_{\beta-p\alpha}$. Then $\mathrm{ad}_f(v) = 0$ and $(\mathrm{ad}_e)^j(v) \neq 0$ for $0 \leq j \leq p + q$. In particular, $(\mathrm{ad}_e)^{p+1}(v) \in \mathfrak{g}_{\alpha+\beta}$ is non-zero. Since $\dim \mathfrak{g}_{\alpha+\beta} = 1$, we immediately have $[\mathfrak{g}_\alpha, \mathfrak{g}_\beta] = \mathfrak{g}_{\alpha+\beta}$.

4. Let $\beta = n\alpha$ with $n \neq 0$. Then $\frac{2(\alpha,\beta)}{(\beta,\beta)} = \frac{2n}{n^2} = \frac{2}{n} \in \mathbb{Z}$, so $n \in \{-2, -1, 1, 2\}$. However, $\mathfrak{g}_{2\alpha} = [\mathfrak{g}_\alpha, \mathfrak{g}_\alpha] = 0$ and $\mathfrak{g}_{-2\alpha} = [\mathfrak{g}_{-\alpha}, \mathfrak{g}_{-\alpha}] = 0$, so $n = \pm 1$.

□

**Theorem 8.6.** *Let $\mathfrak{g}$ be a semisimple complex Lie algebra, and $\mathfrak{h}$ a Cartan subalgebra.*

*(i)* *$\Phi$ spans $\mathfrak{h}^*$.*

*(ii)* *$(\alpha, \beta) \in \mathbb{Q}$ for all $\alpha, \beta \in \Phi$.*

*(iii)* *$(\cdot, \cdot)$ induces a positive definite symmetric bilinear form on $\mathfrak{h}^*_{\mathbb{Q}} \times \mathfrak{h}^*_{\mathbb{Q}}$ with values in $\mathbb{Q}$, where $\mathfrak{h}^*_{\mathbb{Q}}$ denotes the $\mathbb{Q}$-span of $\Phi$.*

*Proof.* (i) If $\Phi$ does not span $\mathfrak{h}^*$, there exists a non-zero $h \in \mathfrak{h}$ such that $\alpha(h) = 0$ for all $\alpha \in \Phi$, implying $[h, \mathfrak{g}_\alpha] = 0$ for all $\alpha$. Since $\mathfrak{h}$ is abelian, $[h, \mathfrak{h}] = 0$ as well, so $h \in Z(\mathfrak{g}) = 0$, a contradiction.

(ii) We have

$$(\alpha, \alpha) = \kappa_{\mathfrak{g}}(t_\alpha, t_\alpha) = \sum_{\gamma \in \Phi} (\gamma, \alpha)^2,$$

so

$$\frac{4}{(\alpha, \alpha)} = \sum_{\gamma \in \Phi} \left( \frac{2(\gamma, \alpha)}{(\alpha, \alpha)} \right)^2 \in \mathbb{Z},$$

implying $(\alpha, \alpha) \in \mathbb{Q}$, so also $(\alpha, \beta) = \frac{1}{2}(\alpha, \alpha) \cdot \frac{2(\alpha,\beta)}{(\alpha,\alpha)} \in \mathbb{Q}$.

(iii) By (ii) we have $(\alpha, \beta) \in \mathbb{Q}$ for $\alpha, \beta \in \mathbb{Q}$, so $(\lambda, \alpha) \in \mathbb{Q}$ for all $\lambda \in \mathfrak{h}^*_{\mathbb{Q}}$ and $\alpha \in \Phi$. Since the Killing form is non-degenerate on $\mathfrak{h}$ and $\Phi$ spans $\mathfrak{h}^*$, the restriction to $\mathfrak{h}^*_{\mathbb{Q}}$ is also non-degenerate. By some above equation, we have $(\lambda, \lambda) \geq 0$ for $\lambda \in \mathfrak{h}^*_{\mathbb{Q}}$, so the form is positive semi-definite. Any non-degenerate positive semi-definite bilinear form is positive definite.

□

# 9 Cartan matrices

**Definition 9.1.** Let $(V, \Phi)$ be a root system and $f : V \to \mathbb{R}$ a linear map such that $f(\alpha) \neq 0$ for all $\alpha \in \Phi$. Then

- $\alpha \in \Phi$ is **positive** if $f(\alpha) > 0$ and **negative** if $f(\alpha) < 0$,
- a positive root $\alpha \in \Phi$ is **simple** if it cannot be written as the sum of two positive roots,
- a **highest root** $\theta \in \Phi$ is a root where $f$ is maximal, that is, $f(\theta) \geq f(\alpha)$ for all $\alpha \in \Phi$.

Denote the set of positive roots by $\Phi_+$, negative roots $\Phi_-$, simple roots $\Pi \subset \Phi_+$.

**Theorem 9.2** (Dynkin)**.** *(i) If $\alpha, \beta \in \Pi$ and $\alpha \neq \beta$, then $\alpha - \beta \notin \Phi$ and $(\alpha, \beta) \leq 0$.*

*(ii) Every positive root is a non-negative integer linear combination of simple roots.*

*(iii) If $\alpha \in \Phi_+ \setminus \Pi$, then $\alpha - \gamma \in \Phi_+$ for some $\gamma \in \Phi$.*

*(iv) ...*

*(v) $\Phi$ is indecomposable if and only if $\Pi$ is indecomposable.*

**Definition 9.3.** Let $\Pi = \{\alpha_1, \ldots, \alpha_r\}$ be the set of simple roots of $\Phi$ (corresponding to $f$). The matrix

$$A = \left( \frac{2(\alpha_i, \alpha_j)}{(\alpha_i, \alpha_i)} \right)_{i,j=1}^{r}$$

is called the **Cartan matrix**.

**Proposition 9.4.** *The Cartan matrix has all entries integers, and satisfies*

*(i) $A_{ii} = 2$ for all $i$,*

*(ii) $A_{ij} \leq 0$ and $A_{ij} = 0 \iff A_{ji} = 0$ for all $i \neq j$,*

*(iii)* $\det A > 0$.

*Proof.* (i) Obvious. (ii) Follows from the above theorem. (iii) Follows from the factorization $A = \operatorname{diag}\left(\frac{2}{(\alpha_i, \alpha_i)}\right)_{i=1}^{r} \cdot ((\alpha_i, \alpha_j))_{i,j=1}^{r}$, and the fact that the Gram matrix of linearly independent vectors is always positive-definite. $\square$

**Remark 9.5.** Any square matrix $A$ satisfying the properties of the above proposition is called an (abstract) **Cartan matrix**.

# 10 Dynkin diagrams

**Example 10.1.** Any $2 \times 2$ Cartan matrix must have the form $A = \begin{pmatrix} 2 & -a \\ -b & 2 \end{pmatrix}$ with $a, b \geq 0$ and $\det(A) = 4 - ab > 0$. Therefore, the only $2 \times 2$ Cartan matrices are the matrices

$$\begin{pmatrix} 2 & 0 \\ 0 & 2 \end{pmatrix}, \quad \begin{pmatrix} 2 & -1 \\ -1 & 2 \end{pmatrix}, \quad \begin{pmatrix} 2 & -1 \\ -2 & 2 \end{pmatrix}, \quad \begin{pmatrix} 2 & -1 \\ -3 & 2 \end{pmatrix},$$

and their transposes.

In particular, note that for any Cartan matrix $A$, the submatrices $\begin{pmatrix} A_{ii} & A_{ij} \\ A_{ji} & A_{jj} \end{pmatrix}$ is also a Cartan matrix.

**Definition 10.2.** The **Dynkin diagram** of an $r \times r$ Cartan matrix $A$ is a graph consisting of $r$ vertices. The edge between vertex $i$ and vertex $j$ is determined by the corresponding $2 \times 2$ matrix:

$$\begin{pmatrix} 2 & 0 \\ 0 & 2 \end{pmatrix} \quad \begin{pmatrix} 2 & -1 \\ -1 & 2 \end{pmatrix} \quad \begin{pmatrix} 2 & -1 \\ -2 & 2 \end{pmatrix} \quad \begin{pmatrix} 2 & -2 \\ -1 & 2 \end{pmatrix} \quad \begin{pmatrix} 2 & -1 \\ -3 & 2 \end{pmatrix} \quad \begin{pmatrix} 2 & -3 \\ -1 & 2 \end{pmatrix}$$

A complete list of connected Dynkin diagrams consists of $D(A_{n\geq 1})$, $D(B_{n\geq 2})$, $D(C_{n\geq 3}), D(D_{n\geq 4})$, $D(E_6)$, $D(E_7)$, $D(E_8)$, $D(F_4)$ and $D(G_2)$, where:

$A_n$

$B_n$

$C_n$

$D_n$

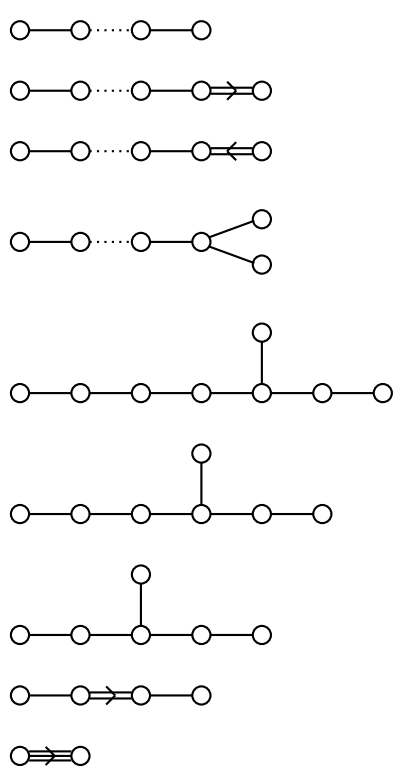
$E_8$

$E_7$

$E_6$

$F_4$

$G_2$



# 11 Classification of semisimple complex Lie algebras

Dynkin diagrams can be used to classify semisimple Lie algebras. Starting from an abstract Cartan matrix, one can construct an associated Lie algebra $\mathfrak{g}(A)$, yielding a circle of constructions:

$$\begin{tikzcd}
\mathfrak{g} \arrow[r, "\text{Cartan } \mathfrak{h}"] & (V = \mathfrak{h}^*_{\mathbb{R}}, \Phi) \arrow[d, "\text{Choose } f \in V^*"] \\
A \arrow[u, "\mathfrak{g}(A)"] & \Pi \arrow[l, "\text{Cartan matrix}"]
\end{tikzcd}$$

**Definition 11.1.** Let $A = (a_{ij})$ be an abstract $r \times r$ Cartan matrix. Let $\tilde{\mathfrak{g}}(A)$ be the Lie algebra generated by $E_i, F_i, H_i$ for $i = 1, \ldots, r$ and satisfying

$$[H_i, H_j] = 0, \quad [H_i, E_j] = a_{ij} E_j, \quad [H_i, F_j] = -a_{ij} F_j, \quad [E_i, F_j] = \delta_{ij} H_i.$$

The generators are called the **Chevallay generators** and the relations the **Chevallay relations**. Note that $\tilde{g}(A)$ is generally infinite-dimensional.

**Lemma 11.2.** *(i) Let $\tilde{\mathfrak{n}}_+ \subset \tilde{g}(A)$ be the subalgebra generated by $E_1, \ldots, E_r$, and $\tilde{\mathfrak{n}}_- \subset \tilde{\mathfrak{g}}(A)$ the subalgebra generated by $F_1, \ldots, F_r$, and $\mathfrak{h}$ the span of $H_1, \ldots, H_r$. Then $\tilde{\mathfrak{g}}(A) = \tilde{\mathfrak{n}}_+ \oplus \mathfrak{h} \oplus \tilde{\mathfrak{n}}_-$ as vector spaces.*

*(ii) $\tilde{\mathfrak{n}}_+ = \bigoplus_{\alpha \in \mathbb{Q}_+} \tilde{\mathfrak{g}}_\alpha$ and $\tilde{\mathfrak{n}}_- = \bigoplus_{\alpha \in \mathbb{Q}_+} \tilde{\mathfrak{g}}_{-\alpha}$ with $\mathbb{Q}_+ =?$.*

*(iii) If $I \subset \tilde{\mathfrak{g}}(A)$ is an ideal, then $I = (\mathfrak{h} \cap I) \oplus (\bigoplus_{\alpha \in \mathbb{Q}_+ \cup -\mathbb{Q}_+} \tilde{\mathfrak{g}}_\alpha \cap I)$.*

*(iv) $\tilde{\mathfrak{g}}(A)$ contains a unique proper maximal ideal $I(A)$ provided $D(A)$ is connected.*

**Definition 11.3.** Let $A = (a_{ij})$ be an abstract $r \times r$ Cartan matrix such that $D(A)$ is connected. Define $\mathfrak{g}(A) = \tilde{\mathfrak{g}}(A)/I(A)$.

**Theorem 11.4.** *(i) Semisimple complex Lie algebras are isomorphic if and only if they have the same Dynkin diagram.*

*(ii) A complete and non-redundant list of simple finite-dimensional complex Lie algebras is the following:*

$$\mathfrak{sl}_n(\mathbb{C}) \ (n \geq 2), \quad \mathfrak{so}_n(\mathbb{C}) \ (n \geq 7), \quad \mathfrak{sp}_{2n}(\mathbb{C}) \ (n \geq 2), \quad E_6, E_7, E_8, F_4 \text{ and } G_2.$$

*Proof.* (i) Let $\mathfrak{g}$ be a simple complex Lie algebra, and $A$ the Cartan matrix corresponding to $\mathfrak{g}$. There is a morphism $\phi : \tilde{\mathfrak{g}}(A) \to \mathfrak{g}$ mapping $E_i \mapsto e_{\alpha_i}, F_i \mapsto f_{\alpha_i}$ and $H_i \mapsto h_{\alpha_i}$. Note that $\phi$ is surjective since the $e_{\alpha_i}, f_{\alpha_i}$ and $h_{\alpha_i}$ generate $\mathfrak{g}$. Indeed, any root $\alpha \in \Phi$ can be written as $\alpha = \sum_{i=1}^r n_i \alpha_i$ for some $n_i \in \mathbb{Z}_{\geq 0}$. Using that $[\mathfrak{g}_\alpha, \mathfrak{g}_\beta] = \mathfrak{g}_{\alpha+\beta}$, one can inductively show that each $\mathfrak{g}_\alpha$ is generated. Now since $\mathfrak{g}$ is simple, the kernel $\ker \phi$ is a maximal proper ideal, so equal to $I(A)$. Therefore, $\phi$ induces an isomorphism $\mathfrak{g}(A) \xrightarrow{\sim} \mathfrak{g}$. This proves the 'if' part. The 'only if' part follows once we show that the Cartan matrix $A$ is independent from $f$. (ii) TODO $\square$

# 12 Representation theory

**Definition 12.1.** Let $\mathfrak{g}$ be a complex semisimple Lie algebra, with Cartan subalgebra $\mathfrak{h}$, root system $\Phi$, and a choice of positive roots $\Phi^+$. An element $\lambda \in \mathfrak{h}^*$ is

- **integral** if $2\frac{(\lambda,\alpha)}{(\alpha,\alpha)}$ is an integer for all $\alpha \in \Phi$,
- **dominant** if $(\lambda, \alpha) \geq 0$ for all $\alpha \in \Phi^+$.

An element $\lambda \in \mathfrak{h}^*$ is **higher** than an element $\mu \in \mathfrak{h}^*$ if $\lambda - \mu$ can be expressed as a linear combination of positive roots with non-negative real coefficients.

**Theorem 12.2** (Theorem of highest weight)**.** *Let $\mathfrak{g}$ be a complex semisimple Lie algebra.*

*(i) Every irreducible finite-dimensional representation of $\mathfrak{g}$ has a unique highest weight, which is dominant and integral.*

*(ii) Two irreducible finite-dimensional representations of $\mathfrak{g}$ with the same highest weight are isomorphic.*

*(iii) If $\mu$ is a dominant integral element, then there exists an irreducible finite-dimensional representation of $\mathfrak{g}$ with highest weight $\mu$.*

*Proof.* (i) Supposed to be easy.

(ii) Supposed to be easy.

(iii) Supposed to be hard. Can be done using Verma modules.

□